US Green Energy Transition: Visualizing the Past, Present and Future

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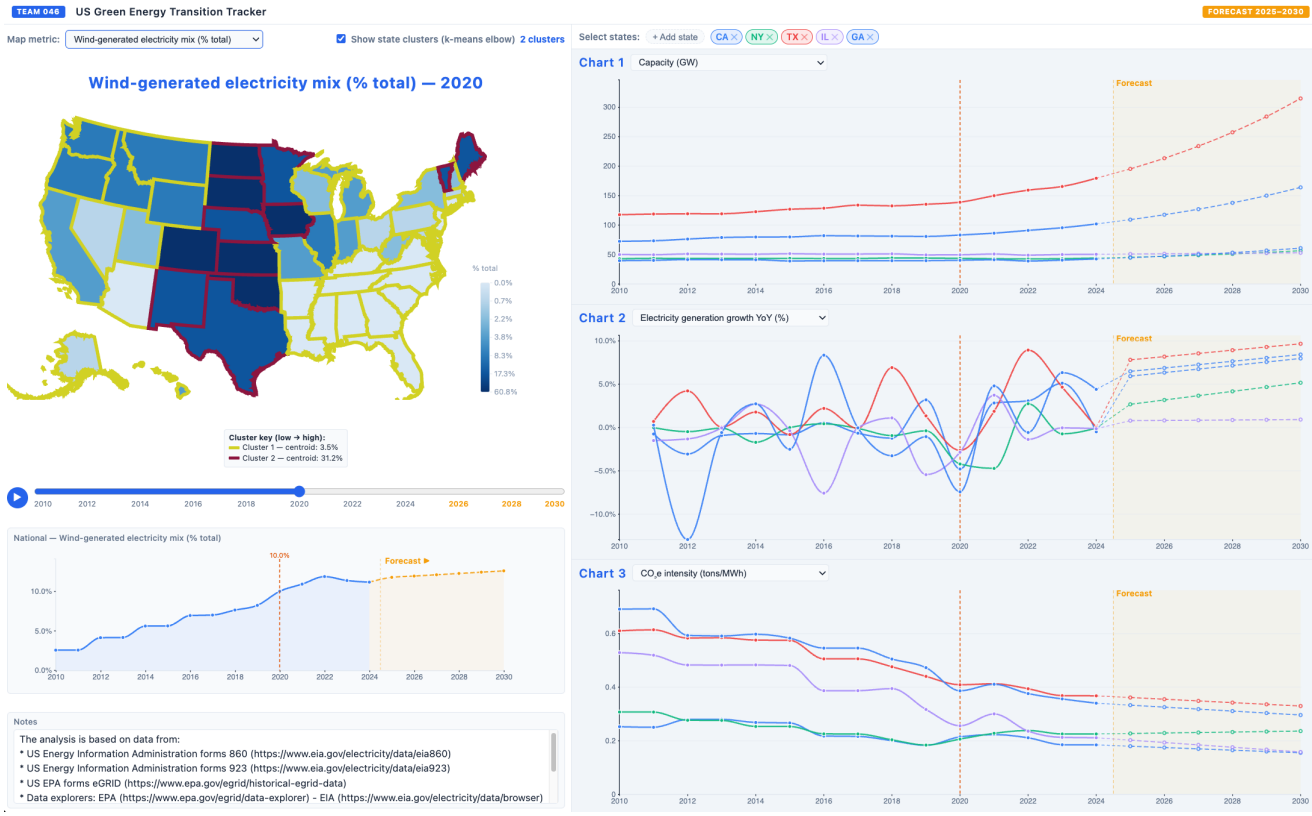


Introduction

We analyze the **US shift towards green energy from 2010 through 2030** based on datasets by the US Energy Information Administration (EIA's forms 860 and 923) and the Environmental Protection Agency (EPA's EGRID). The data contains detailed information about the c30K energy generators in the US, such as fuel types, energy generation and capacity, as well as CO2, NOx and SOx emissions, among others. **We analyze the change in the energy generation mix** by fuel type, the trajectory of emissions, cluster states by energy metrics and forecast all of these through 2030 under a baseline scenario. **We visualize our analysis in an interactive dashboard**.

Problem Definition

The US energy sector is undergoing a profound transformation, with fossil fuel generators transitioning out of service and gigawatts of renewable capacity coming online nationwide. Despite the existence of an entire government agency dedicated to the collection and analysis of energy data (EIA, and partly EPA), there is no widely available tool that combines historical analysis and forward-looking forecasts in a narrative-driven user experience. The two leading dashboards / data explorers illustrate the "pain point" for decision makers: the EIA's Electricity Data Browser is query-driven and presents raw time series with little analytics and forecasts, while EPA's eGRID Data Explorer offers no visualization to begin with. Companies such as Carbon Brief produce excellent thematic graphics but do not support user-driven exploration or forecasts. As a result, decision-

makers - utilities, states, asset managers, NGOs, and citizens - must either pull together data manually from various sources or pay for the services of a data analytics firm. Our project addresses this gap by delivering an interactive, narrative-driven dashboard that integrates 15 years of historical electricity generation, capacity, fuel consumption and emissions data with state-level forecasts to 2030, augmented by a k-means algorithm to "unearth" state clusters. The end result is a single tool that lets a user move from a national overview to a state-level deep-dive, to a forecast, in a matter of clicks, thus closing the gap between the EIA and EPA's data explorers and commercially available analytical tools.

Literature Survey

In terms of existing energy analytical methodologies, Wang et al [9] assign approaches to four main groups: preprocessing (e.g., fuzzy sets, k-means cluster), deep learning (e.g., DBN, GAN), optimization (e.g., external, trial-and-error) and regression (e.g., linear, logistic; SVM, FNN, LSTM). Kim et al [5] apply time series clustering to smart grid data while Al-Lawati et al [6] apply stochastic optimization to EIA data. Bouziane et al [8] apply LTSM and Monte Carlo simulations to predict energy generation and model uncertainty. An important takeaway is the use of clustering, with Sim et al [7] concluding that it meaningfully improves forecasting accuracy. Mustafa et al [10] review 145 published articles on the use of time series to forecast energy data and conclude that classical approaches (ARIMA, exponential smoothing) are best suited for "linear or regularly seasonal situations", with gray, fuzzy and machine learning (e.g., XGBoost) models better suited for other situations.

In terms of existing visualization approaches, the EIA's Electricity Data Browser has basic charting capabilities while the EPA's eGRID does not have visualization tools. CarbonBrief has state-of-the-art infographics and charts on US electricity generation but no forecasting capabilities. Chen et al [1] provide a review of various screen-, computer- and app-based visualization methods for energy data: maps (dot-based, line-based and area-based GIS maps), 3D models, animation and VR/AR. Cook et al [3] discuss a power flow model utilizing the EIA-860 dataset to visualize crucial information about US electricity generation in a series of line charts and maps. Kunkolienkar et al [4] develop an interactive dashboard which is used to visualize days with similar weather and the corresponding renewable energy power generation.

In general, existing approaches to analyzing and visualizing energy data suffer from at least one of the following three challenges: 1) visualizations are query-driven (not enough of a narrative); 2) the analytical layer is simplistic and provides little informational value beyond the obvious; and 3) the analysis is backward-looking. We agree with Molnar & al who argue in [2] that the complexity of energy data makes it hard to communicate effectively to stakeholders.

Our project presents a novel combination of a narrative-driven, interactive visualization with a non-trivial analytical layer. The analytical layer models the change in the energy mix over time and energy transition 'state clusters'; incorporating the importance of politics and green policies will be a valuable feature to develop in the future [14], [15], [16], [17]. Importantly, our analytical layer is forward-looking to 2030. The over-arching goal of our project is to improve decision making for **the stakeholders that care: corporations, NGOs, asset managers / owners, real estate / operators, governments (federal, state, county) and individuals.** Bonatto et al [18] discuss an application of big data analytics for energy sustainability for the Brazilian public sector while Stecyk et al [19] examine how real-time data and visualization tools

can help energy planners interpret complex information quickly and efficiently. Our goals are very much the same.

Proposed Method

Our methodology is straightforward. First, we obtained the raw data in XLS files from the EIA and EPA websites. We then wrangled, cleaned and joined the data by downcasting the plant-level EPA data to the generator-level EIA data. The result was 411K rows of data with 45 columns across 15 years (107GB). Loaded as a csv file, this is the data which powers our visualization.

Figure 1. Raw data collected from EPA and EIA

Common name	Unit	Name	Form	File	Sheet	Dimension
nameplate capacity	MW	capacity	860	3_1_Generator	Operable	generator
			egrid	egrid2024_data_rev2	PLNT24	plant
electricity net generation	MWh	net_elec_gen	923	EIA923_Schedules_2_3_4_5_M_12_2023_Final_Revision	Page 1 Generation and Fuel data	by plant & fuel type
fuel heat - *by fuel type*	MMBtu	fuel_heat_tot / fuel_heat_tot_[TYPE]				
electricity generation - *by fuel type*	MWh	net_elec_gen / net_elec_gen_[TYPE]	egrid	egrid2024_data_rev2	PLNT24	plant
NOX emissions	tons	emis_nox	egrid	egrid2024_data_rev2	PLNT24	plant
SO2 emissions	tons	emis_so2	egrid	egrid2024_data_rev2	PLNT24	plant
CO2 emissions	tons	emis_co2	egrid	egrid2024_data_rev2	PLNT24	plant
CH4 emissions	lbs	emis_ch4	egrid	egrid2024_data_rev2	PLNT24	plant
N2O emissions	lbs	emis_n2o	egrid	egrid2024_data_rev2	PLNT24	plant
CO2e emissions	tons	emis_co2_equiv	egrid	egrid2024_data_rev2	PLNT24	plant

The visualization runs on D3.js, with state and national aggregation taking place dynamically (so that new data can be added in the future). Once the aggregation is complete, we calculate the following metrics:

Figure 2. Calculated metrics

Common name	Unit	Name	Formula
growth rates	% yoy	[METRIC]_pct_yoy	capacity, fuel_heat_tot, net elec gen, net elec gen by fuel type, emissions (NOX, SO2, CO2, CH4, N2O, CO2e)

% electricity mix by fuel type consumed	% of total	net_elec_gen_[TYPE]_pct	net_elec_gen_[TYPE] / net_elec_gen
capacity utilization	%	capacity_util_rate	net_elec_gen / (capacity * 8760)
heat rate	MMBtu / MWh	heat_rate_tot	fuel_heat_tot / net_elec_gen
emissions rate	tons / MMBtu	emissions_rate	emis_co2_equiv / fuel_heat_tot
NOX emissions rate	tons / MMBtu	emissions_rate_nox	emis_nox / fuel_heat_tot
SO2 emissions rate	tons / MMBtu	emissions_rate_so2	emis_so2 / fuel_heat_tot
CO2 emissions rate	tons / MMBtu	emissions_rate_co2	emis_co2 / fuel_heat_tot
CH4 emissions rate	lbs / MMBtu	emissions_rate_ch4	emis_ch4 / fuel_heat_tot
N2O emissions rate	lbs / MMBtu	emissions_rate_n20	emis_n2o / fuel_heat_tot

The algorithm then forecasts most of the above variables using the 5-year average trend with 0.5 damping factor and 0 floors where relevant to prevent explosive forecasts. The remaining metrics are then calculated as follows:

Figure 3. Forecasting approach for derived metrics

Metric / variable	**Forecating formula**
capacity at T	[capacity at T-1] x [capacity_pct_yoy at T]
net_elec_gen at T	[capacity at T] x [capacity_util_rate at T]
net_elec_gen_[TYPE] at T	[net_elec_gen at T] x [net_elec_gen_[TYPE]_pct]
fuel_heat_tot at T	[net_elec_gen at T] x [heat_rate_tot at T]
emis_co2_equiv at T	[fuel_heat_tot at T] x [emissions_rate at T]
emis_nox at T	[fuel_heat_tot at T] x [emissions_rate_nox at T]
emis_so2 at T	[fuel_heat_tot at T] x [emissions_rate_so2 at T]
emis_co2 at T	[fuel_heat_tot at T] x [emissions_rate_co2 at T]
emis_ch4 at T	[fuel_heat_tot at T] x [emissions_rate_ch4 at T]
emis_n2o at T	[fuel_heat_tot at T] x [emissions_rate_n2o at T]

Lastly, we perform simple but powerful clustering which groups states into two clusters, depending on their distance from the two centroids. The clusters change every year and are useful for visualizing the transition journey of each state given an energy metric. The final dashboard front-end consists of a US choropleth, three configurable state-specific charts, a multi-state pill selector with select-all / unselect-all, and a notes panel — all driven by D3.js.

Evaluation

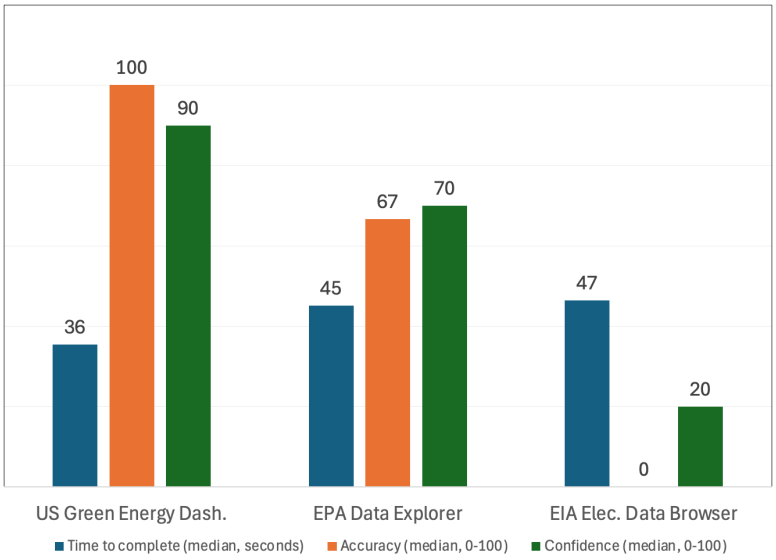
There were four stages to the **analytical process**: 1) data wrangling and cleaning; 2) calculation of metrics; 3) forecasting; and 4) clustering. We sanity-checked wrangled/cleaned data and calculated metrics by visualizing them across states and years. We cross-checked forecasts against studies available in the public domain (for example, capacity utilization rate is in the 35-40%). We also used domain experts to sanity-check all of our results, including

clustering. Other than the data cleanup, which took a considerable amount of time, the outcomes of the sanity- and cross-checks confirmed the veracity of the data and the analysis.

As for **evaluation of the visualization**, our review of [11], [12] and [13] shows that ground-truth data, systematic benchmarking and statistical measures (prediction error, bias) are well-established approaches. We performed a task-based usability evaluation (5 users, 6 tasks, quantitative metrics), system usability scales based on scored answers, and qualitative comments from the users. Our five users were assigned six tasks to perform using our team's dashboard (https://calm-youtiao-4824a3.netlify.app) as well as the EIA Data Browser (https://www.eia.gov/electricity/data/browser) and the EPA Data Explorer (https://www.epa.gov/egrid/data-explorer). Quantitative and qualitative feedback was used to measure the effectiveness and system usability of the team's dashboard.

Figure 4 below benchmarks the results across the three dashboard solutions: our solution enabled the fastest median response time of 36 seconds per question, 9 seconds better than the EPA and 11 seconds better than the EIA. From 0-100, the median accuracy of our users' responses was an impressive 100 vs EPA at 67 and EIA at 0 (most users gave up on the EIA given the relative difficulty of navigating the dashboard). In terms of confidence of 0-100, our solution garnered 90 vs 70 for the EPA and 20 for the EIA. Interestingly, on task 6 which has to do with forecasting, users achieved 67 both using our tool and the EPA, but 0 on EIA: yet users reported confidence of 86 on our solution compared to 26 for the EPA and 20 for the EIA.

Figure 4. Results of user testing - 6 tasks to complete on 3 dashboards



Tasks for all 3 dashboards

1) What was TX's electricity generation in 2022?
2) Which state emitted most CO_2 in 2020?
3) How has California's coal-fired generation changed from 2010 to 2023 up, down, or flat?
4) What share (%) of New York's electricity came from nuclear in 2023?
5) Compare wind generation in Iowa and Kansas in 2023 — which is higher?
6) What's a reasonable estimate for US CO_2e emissions in 2030?

We also calculated system usability scales for the 3 dashboards based on answers from our users. As Figure 5 demonstrates, our solution scored 86 points (A-grade equivalent) compared to 54 for EPA ("D") and 33 for EIA ("F").

Figure 5. System Usability Scales for the 3 Dashboards

Statement	US Green Energy Dash.	EPA Data Browser	EIA Elec. Data Explorer
I think I would like to use this tool frequently.	4.00	2.80	2.00
I found this tool unnecessarily complex.	1.40	3.00	4.20
I thought this tool was easy to use.	4.40	2.80	2.00
I think I would need help to use this tool.	1.60	2.80	3.00
I found the various functions in this tool well integrated.	4.20	3.20	2.40
I thought there was too much inconsistency in this tool.	2.00	2.20	3.60
I would imagine most people would learn to use this tool quickly.	4.60	3.00	2.60
I found this tool very awkward to use.	1.20	2.40	4.00
I felt very confident using this tool.	4.40	2.80	1.80
I needed to learn a lot before I could get going with this tool.	1.20	2.80	3.00
System usability scale	**85.50**	**53.50**	**32.50**
Grade-equivalent	**A**	**D**	**F**

Lastly, qualitative answers from the users to five additional questions confirmed their strong preference for our solution compared to the other two. The questions we asked were: 1) Which tool was easiest and why? 2) Which gave you the most trustworthy answer and why? 3) What was missing in our dashboard that you wished was there? 4) What was confusing or annoying in our dashboard?; and 5) Would you use our dashboard for real work / a class project - why or why not?). Details of the answers to the questions are available from the authors of this report.

Conclusions and Discussion

We achieved our main goal: to create a narrative-driven visualization that combines historical analysis, forward-looking forecasts and an interactive user experience in a single interface. Our dashboard ingests 411K rows of data over 15 years, computes 29 metrics (e.g., capacity utilization, % mix of electricity, emissions rates/intensities) and projects each one to 2030. The dashboard front-end consists of a US choropleth, three configurable state-specific charts, a multi-state pill selector with select-all / unselect-all, and a notes panel — all driven by D3.js.

The successful conclusion of our project should add significant value to stakeholders: the ability to synthesize complex energy data into a coherent narrative and actionable insights; interactive visualizations; and a solid forecasting approach. Our work should help identify green-energy champion vs laggard states, uncover "hidden" trends and allow users to plan for the future - which is valuable given the discontinuous nature of green energy policies. The results of our project should be able to inform policy and capital allocation decisions at the local and state levels.

In terms of limitations and future improvements, our forecasts use a damped linear trend on the last 5 historical years; this is a conservative choice to avoid explosive forecasts but does not capture discontinuous events such as policy shifts and future plant retirements and additions (the visualization does not contain explicit gap analysis). The state clusters could be re-run to become more meaningful by incorporating multiple variables in the k-means algorithms. "Green" policies can be included in the future as an overlay. And lastly, the introduction of user-defined scenarios would make our solution even more valuable to stakeholders.

Effort statement: All team members have contributed a similar amount of effort.

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